



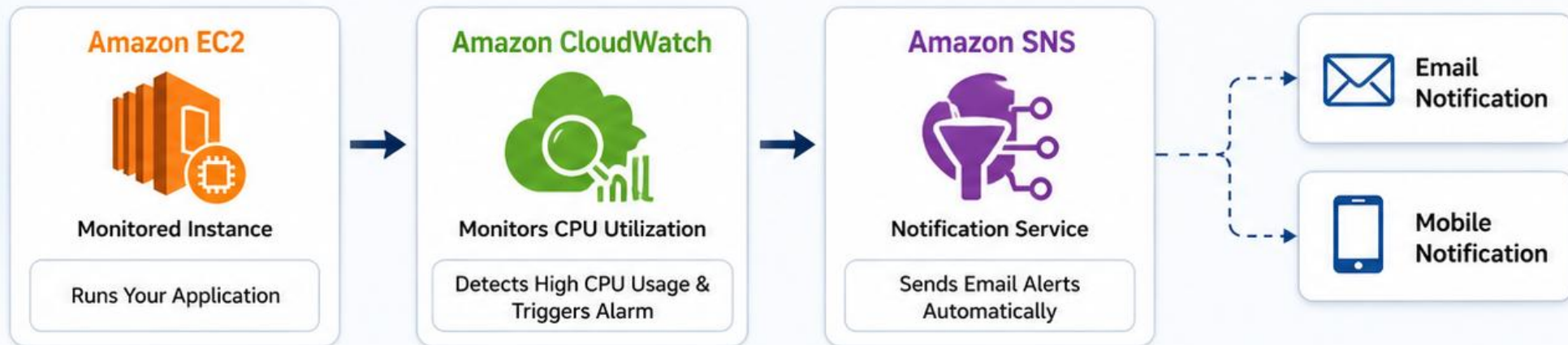
A PROJECT REPORT ON



AWS CLOUDWATCH ALARM & AMAZON SNS NOTIFICATION MANAGEMENT

• TO MONITOR AN AWS EC2 INSTANCE •

Real-Time CPU Utilization Monitoring & Automated Email Notifications
using AWS CloudWatch and Amazon SNS



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Project Introduction

Imagine you own a large shopping mall. Since you can't watch every corner of the mall all the time, you hire a **security guard** to keep an eye on everything.

As long as everything is normal,

But if there is a fire, or any emergency, the guard immediately calls the manager so that quick action can be taken before the situation becomes serious.

Now think about a computer server in the same way.

A server also needs someone to monitor its health **24/7**.

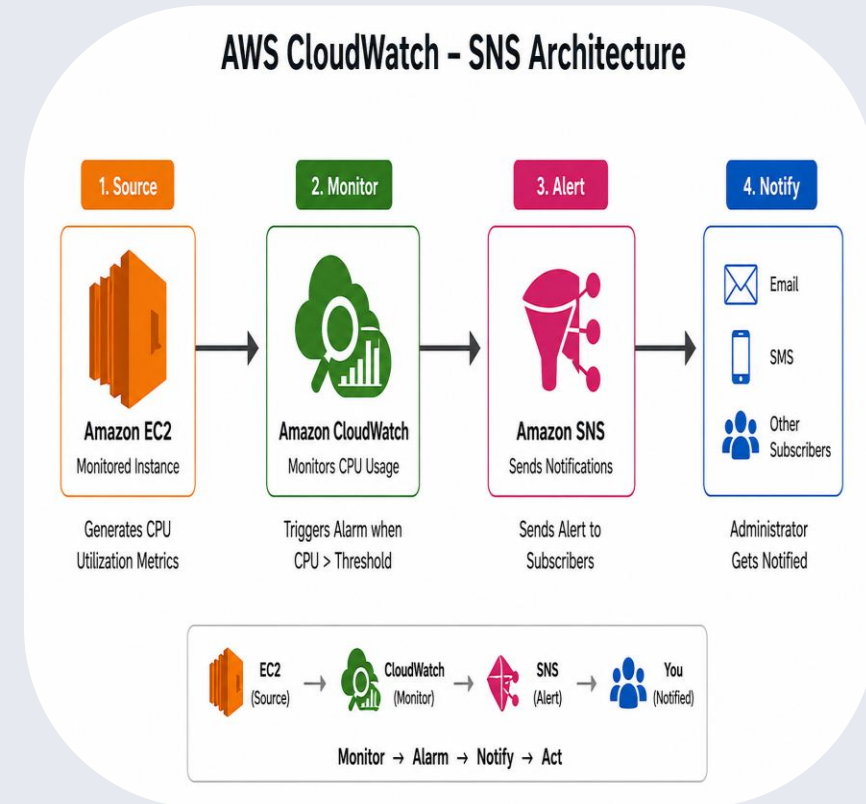
If its CPU usage becomes too high and nobody notices, the application may become slow or even stop working.

That's exactly the problem my project solves.

In my project :

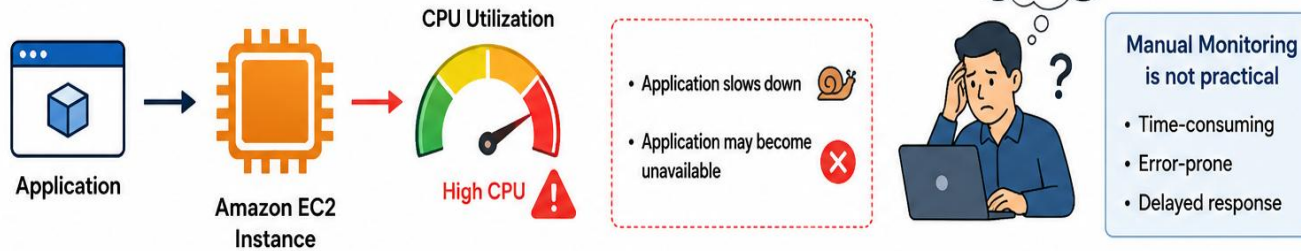
- Amazon EC2 - is the server where the application is running.
- Amazon CloudWatch - acts like the security guard. It continuously monitors the server's CPU usage. If the CPU usage crosses the defined threshold, CloudWatch immediately triggers an alarm.
- Amazon SNS - acts like the phone call. It instantly sends an email notification to the administrator.

The administrator receives the alert and takes quick action before the problem affects users. This is how my project provides **real-time monitoring and automated notifications**, helping improve system reliability and reduce downtime.

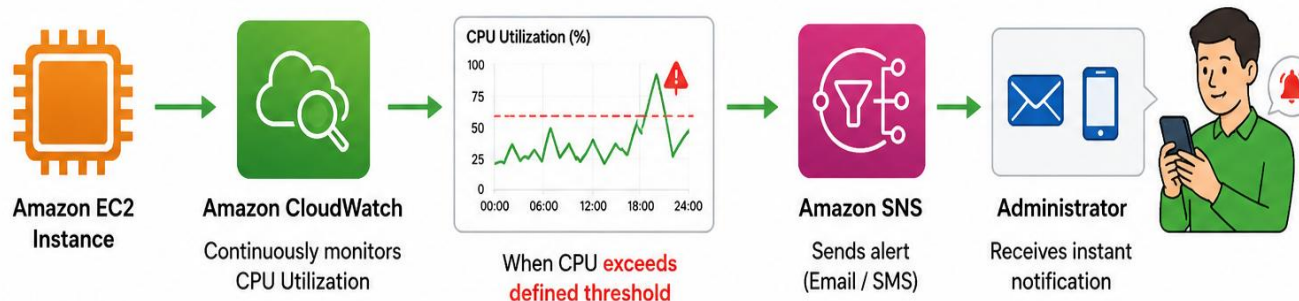


Why Monitoring Is Important ?

THE PROBLEM



THE SOLUTION



Quick alerts enable quick action and prevent downtime before it affects users.

Problem Statement

1. High CPU usage can reduce application performance
2. Manual server monitoring is time-consuming
3. Delayed alerts
4. Downtime affects users and business operations.

Solution Statement

1. Deploy the application on an **Amazon EC2** instance.
2. Use **Amazon CloudWatch** to continuously monitor CPU utilization.
3. Configure a **CloudWatch Alarm** with a predefined CPU threshold.
4. Integrate the alarm with **Amazon SNS**.
5. Automatically send **email notifications**

Which AWS Services Used In My Project

➤ This project uses three core AWS services.

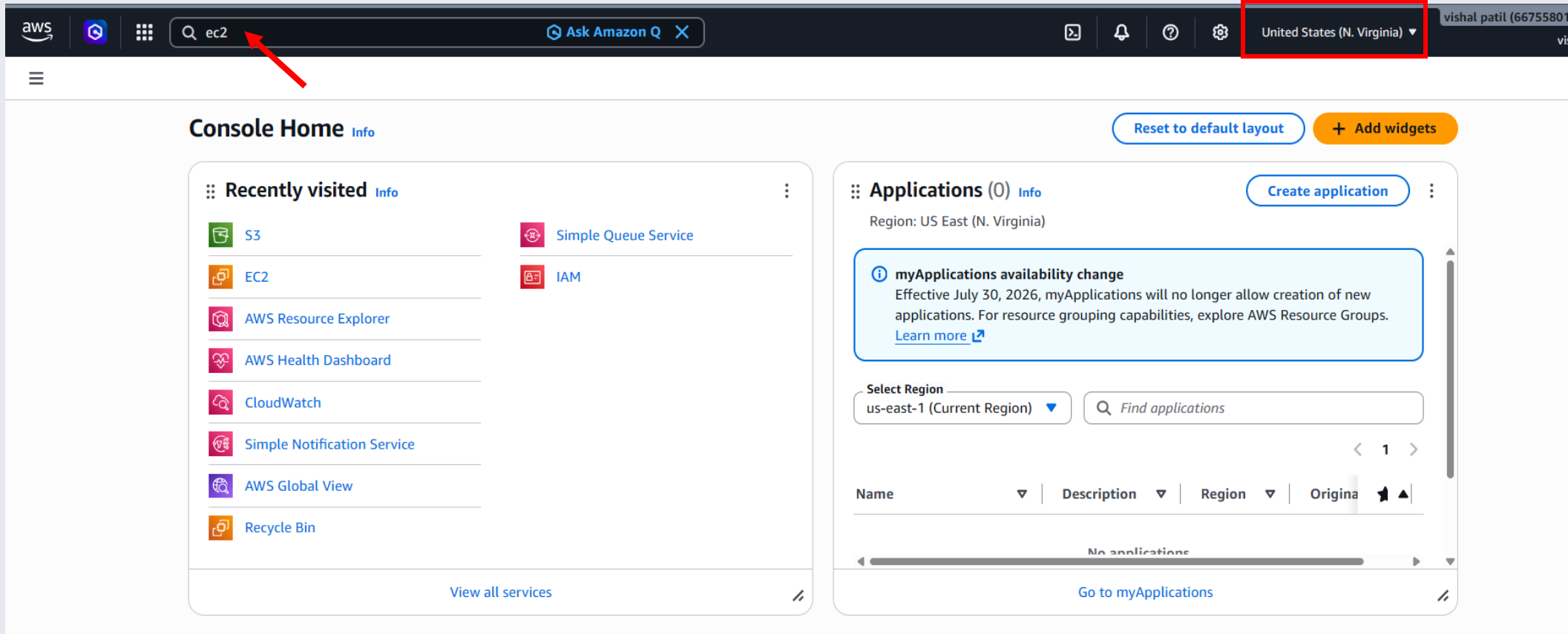
1. Amazon EC2 (Elastic Compute Cloud)
2. Amazon CloudWatch
3. Amazon SNS (Simple Notification Service)



AWS Service	What It Does
Amazon EC2	Runs the server/application
Amazon CloudWatch	Monitors CPU usage
Amazon SNS	Sends email alerts

Implementation Steps

- ❑ Open AWS Console Management and Search EC2



1. Log in to AWS
2. Select the region
3. Search EC2

☐ Launch Instance

The screenshot shows the AWS Management Console interface for the 'Launch Instance' page. The top navigation bar includes the AWS logo, a search bar, and user information. The left sidebar shows the 'EC2' menu with options like 'Dashboard', 'AWS Global View', 'Events', and 'Instances'. The main content area has a dark header with 'Amazon Elastic Compute Cloud (EC2)' and a sub-header 'Create, manage, and monitor virtual servers in the cloud.' Below this, a description states that Amazon EC2 offers the broadest and deepest compute platform. On the right, there are two buttons: 'Launch instance' (highlighted with a red box) and 'View dashboard'. Below these buttons is a section titled 'Additional actions'.

aws [Search] [Alt+S] United States (N. Virginia) vishal patil (6675580127)

EC2

- Dashboard
- AWS Global View
- Events
- ▼ Instances
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts

Compute

Amazon Elastic Compute Cloud (EC2)

Create, manage, and monitor virtual servers in the cloud.

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 600 instance types and a choice of the latest processors, storage, networking, operating systems, and purchase models to help you best match the needs of your workload.

Launch a virtual server

Launch instance

[View dashboard](#)

Additional actions

☐ Give the name of Instance

The screenshot shows the 'Name and tags' section of the AWS console. The 'Name' field is highlighted with a red box and contains the text 'cloud-watch-SNS-110'. To the right of the field is a button labeled 'Add additional tags'. Below this section is the 'Application and OS Images (Amazon Machine Image)' section, which includes a search bar and a list of operating systems under the 'Quick Start' tab. The list includes Amazon Linux, macOS, Ubuntu, Windows, Red Hat, SUSE Linux, and Debian.

Name and tags [Info](#)

Name

cloud-watch-SNS-110 [Add additional tags](#)

▼ **Application and OS Images (Amazon Machine Image)** [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the [Browse more AMIs](#).

[Search our full catalog including 1000s of application and OS images](#)

Recents **Quick Start**

- Amazon Linux
- macOS
- Ubuntu
- Windows
- Red Hat
- SUSE Linux
- Debian

- **Instance Type** – select t3 micro for free tier
- **Key pair** – proceed without key pair

▼ **Instance type** [Info](#) | [Get advice](#)

Instance type

t3.micro

Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0104 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
On-Demand SUSE base pricing: 0.0104 USD per Hour On-Demand RHEL base pricing: 0.0392 USD per Hour

☐ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

▼ **Key pair (login)** [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

Proceed without a key pair (Not recommended)

Default value ▼

[↻ Create new key pair](#)

Create billing usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.

Create billing alerts

Connect to your instance

Once your instance is running, log into it from your local computer.

Connect to instance

Learn more

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

Connect an RDS database

Create a new RDS database

Learn more

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

Create EBS snapshot policy

Manage detailed monitoring

Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute period.

Manage detailed monitoring

Create Load Balancer

Create an application, network gateway or classic Elastic Load Balancer

Create Load Balancer

Create AWS budget

AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a single location.

Create AWS budget

Manage CloudWatch alarms

Create or update Amazon CloudWatch alarms for the instance.

Manage CloudWatch alarms

View all instances

- After create instance then click on view all instance

Instances (1/1) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...
☒	cloud-watch-S...	i-0dbba5d2558e8c59e	Running	t3.micro	Initializing	us-east-1d	ec2-54-211-253-67.co...	54.211.253.67

Launch instances

Launch instance from template

Migrate a server

Connect

Stop instance

Start instance

Reboot instance

Hibernate instance

Terminate (delete) instance

Instance diagnostics

Instance settings

Networking

Security

Image and templates

Storage

Monitor and troubleshoot

Manage detailed monitoring

Manage CloudWatch alarms

Configure CloudWatch agent

Get system log

Get instance screenshot

Connect to serial port

Replace root volume

Visit SSM Fleet Manager

i-0dbba5d2558e8c59e (cloud-watch-S

Details

Status and alarms

Mo

Instance summary Info

Instance ID

i-0dbba5d2558e8c59e

IPv6 address

Private IPv4 addresses

172.31.41.130

Public DNS

- Right click on instance
- Click on monitor and troubleshoot
- Select manage cloudwatch alarms

Alarm notification [Info](#)

Configure the alarm to send notifications to an Amazon SNS topic when it is triggered.

cloud-watch-SNS

- Set cloudwatch name

Alarm thresholds

Specify the metric thresholds for the alarm.

Group samples by

Average

Alarm when

>=

Consecutive period

1

Alarm name

awsec2-i-0dbba5d2558e8c59e-GreaterThanOrEqualToThreshold-CPUUtilization

Alarm description

Alarm on instance i-0dbba5d2558e8c59e: Triggered when CPUUtilization >= 0.99 for 1 consecutive 5-minute periods.

Type of data to sample

CPU utilization

Percent

40

Period

5 Minutes

- Set percentage

Sample metric data [Info](#)

Sample metric data for i-0dbba5d2558e8c59e

☐ Alarm recommendations 

[Investigate with AI - new](#)

1h

3h

12h

1d

3d

1w

Custom 

UTC timezone 

[Explore related](#)



CPU utilization (Average)

Percent

40% 40

20%

0%

13:40

13:50

14:00

14:10

14:20

14:30

14:40

14:50

15:00

15:10

15:20

15:30

15:40

15:50

16:00

16:10

16:20

16:30

 CPUUtilization

Cancel

Create

aws    [Ask Amazon Q](#) 

[EC2](#) > [Instances](#)

EC2

- Dashboard
- AWS Global View 
- Events
- ▼ **Instances**
 - [Instances](#)
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
 - Capacity Manager [New](#)
- ▼ **Images**
 - AMIs
 - AMI Catalog

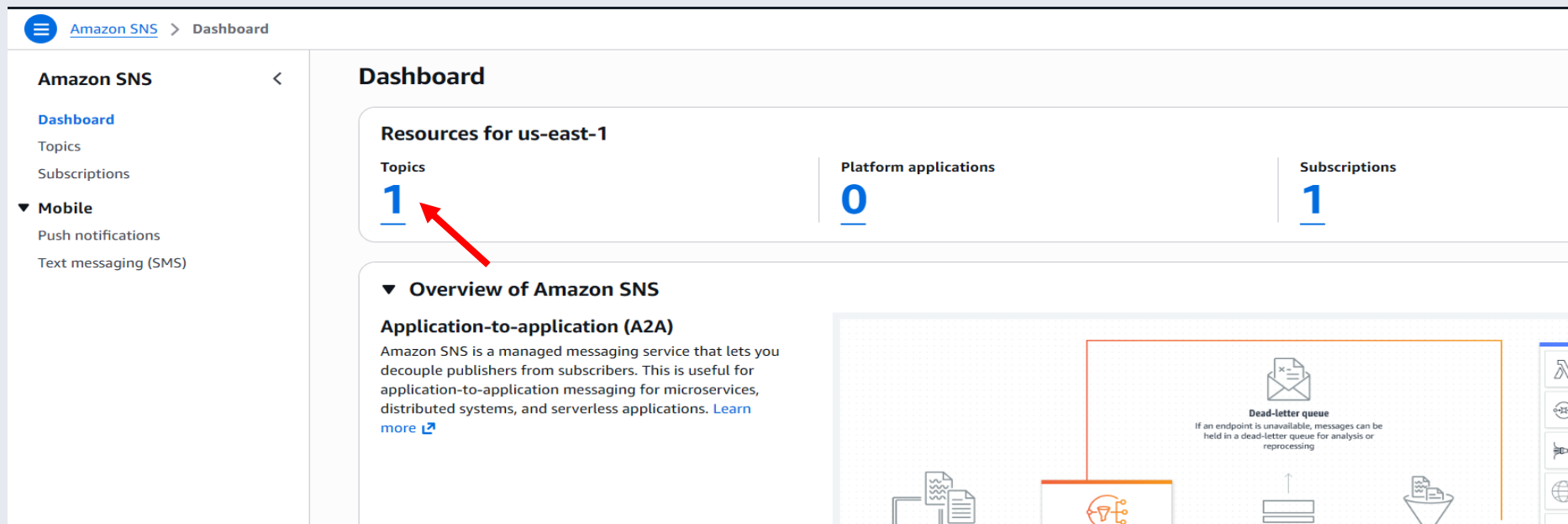
Services [Show more](#)

-  **Simple Notification Service**
SNS managed message topics for Pub/Sub 
-  **AWS Sustainability**
Track, measure, and review the environmental impact from your AWS usage 
-  **Route 53 Resolver**
Resolve DNS queries in your Amazon VPC and on-premises network.

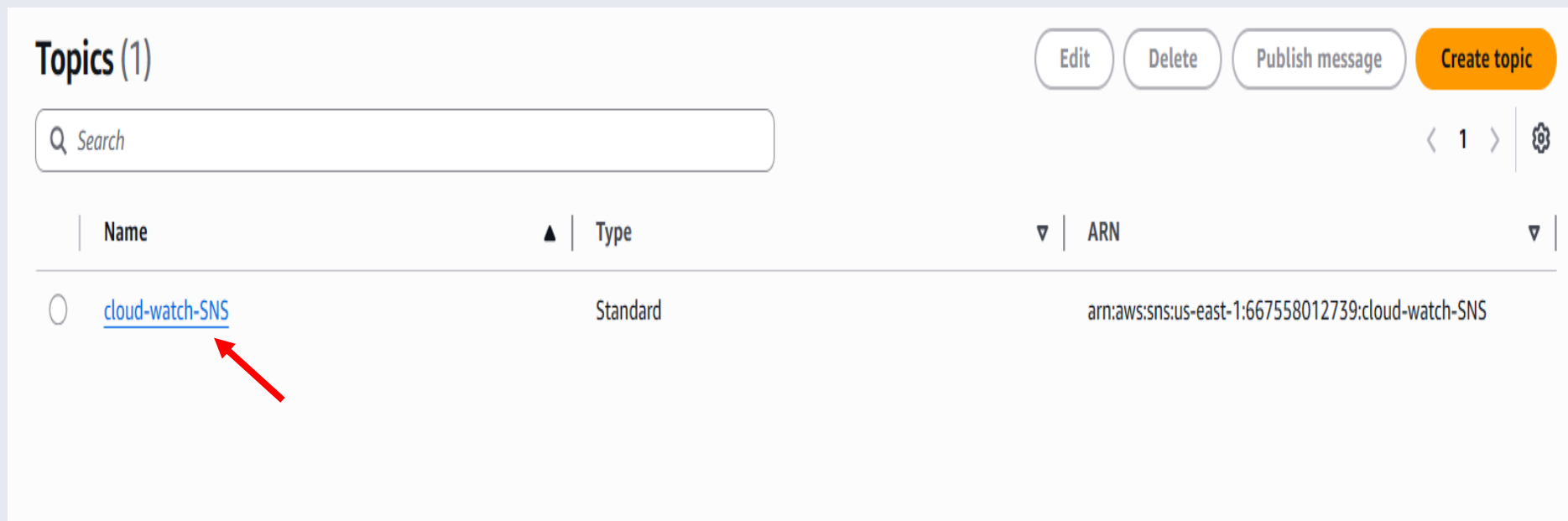
Features [Show more](#)

- Events**
 -  ElastiCache feature
- SMS**
 -  AWS End User Messaging feature

- Back to EC2 instance Tab
and search SNS



- After that interface click on Topics



- Click on cloud-watch-SNS

Amazon SNS > Subscriptions

Amazon SNS

Dashboard

Topics

Subscriptions

▼ Mobile

Push notifications

Text messaging (SMS)

New Feature
Amazon SNS now supports High Throughput FIFO topics. [Learn more](#)

Subscriptions (0) [Edit](#) [Delete](#) [Request confirmation](#) [Confirm subscription](#) [Create subscription](#)

Search

< 1 > ⚙

ID	Endpoint	Status	Protocol	Topic
No subscriptions found				

[Create subscription](#)

- Click on Create Subscription Button

Details

Topic ARN
arn:aws:sns:us-east-1:667558012739:cloud-watch-SNS

Protocol
The type of endpoint to subscribe
Email

Endpoint
An email address that can receive notifications from Amazon SNS
vp228459@gmail.com

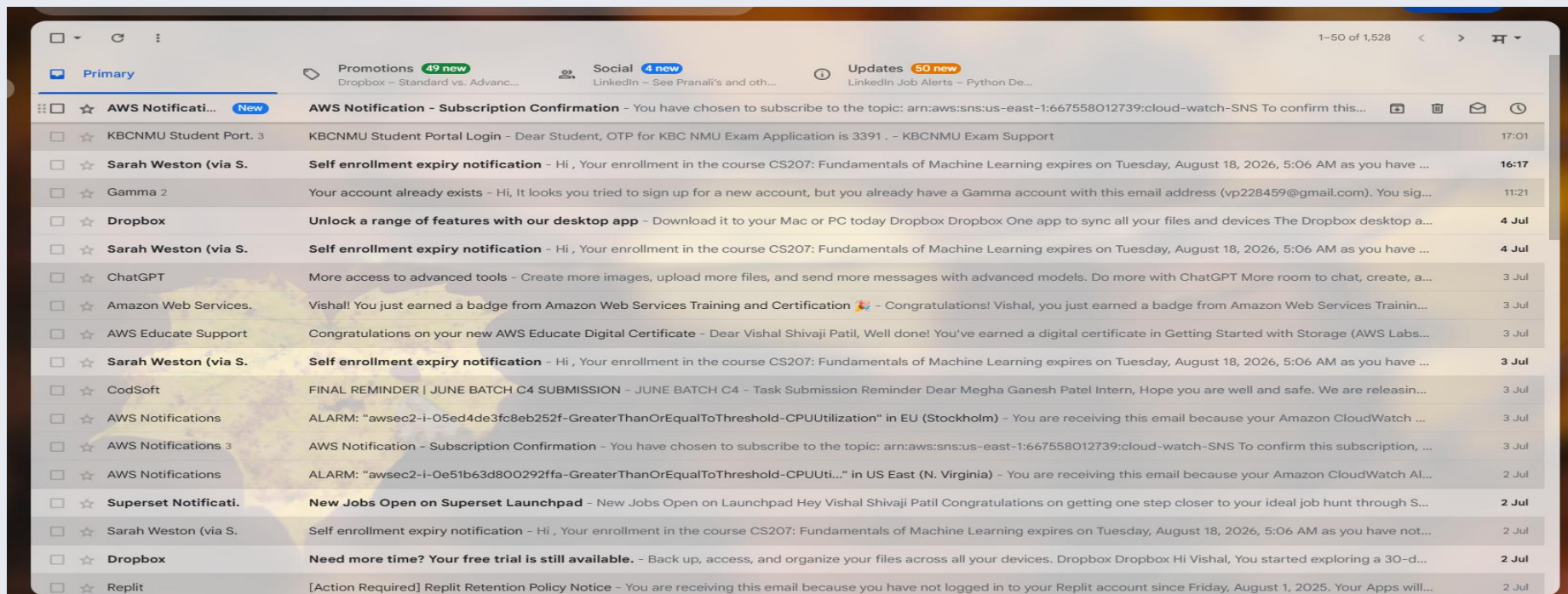
After your subscription is created, you must confirm it. [Info](#)

► **Subscription filter policy - optional** [Info](#)
This policy filters the messages that a subscriber receives.

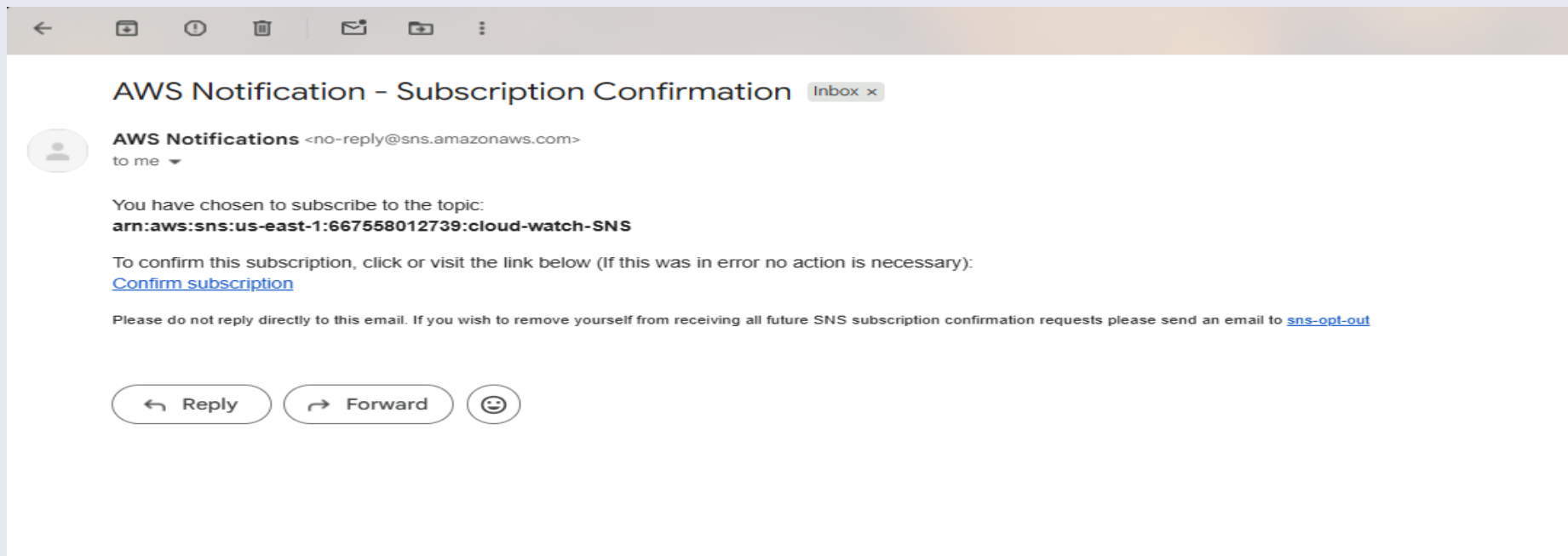
► **Redrive policy (dead-letter queue) - optional** [Info](#)
Send undeliverable messages to a dead-letter queue.

[Cancel](#) [Create subscription](#)

- Select protocol as a Email
- Put email on Endpoint
- Create Subscription



- Open Gmail
- Click on AWS notification



- Click on confirm subscription

Subscription: c6170bd7-17f0-464c-97f0-2846be054234

Edit

Delete

Details

ARN

 arn:aws:sns:us-east-1:667558012739:cloud-watch-SNS:c6170bd7-17f0-464c-97f0-2846be054234

Endpoint

vp228459@gmail.com

Topic

[cloud-watch-SNS](#)

Status

 Confirmed 

Protocol

EMAIL



✓ Created CloudWatch alarm: awsec2-i-0425e18936d3af5ef-GreaterThanOrEqualToThreshold-CPUUtilization



Instances (1/1) [Info](#)

Last updated
1 minute ago



Connect

Instance state ▼

Actions ▼

Launch instances ▼

Saved filter sets

Choose filter set ▼

Find Instance by attribute or tag (case-sensitive)

< 1 >



☒	Name	Instance ID	Instance state ▼	Instance type ▼	Status check	Availability Zone ▼	Public IPv4 DNS ▼	Public IPv4 ... ▼
☒	cloud-watch-s...	i-0425e18936d3af5ef	✓ Running	t3.micro	✓ 3/3 checks passed	us-east-1c	ec2-100-26-61-11.com...	100.26.61.11

Back to EC2 instance and
connect the instance

Instances (1/1) [Info](#)

[Connect](#)[Instance state](#) ▼[Actions](#) ▼[Launch instances](#) ▼

Saved filter sets

[Choose filter set](#) ▼

< 1 >



☒	Name ✎	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...
☒	cloud-watch-S...	i-0dbba5d2558e8c59e	Running 🔍 🔍	t3.micro	3/3 checks passed	us-east-1d	ec2-54-211-253-67.co...	54.211.253.67

i-0dbba5d2558e8c59e (cloud-watch-SNS-110)

[Details](#)[Status and alarms](#)[Monitoring](#)[Security](#)[Networking](#)[Storage](#)[Tags](#)☒ Include metrics in the CWAgent namespace [?](#)[Learn more](#) [🔗](#)☐ Alarm recommendations [?](#)[Investigate with AI - new](#)[1h](#)[3h](#)[12h](#)[1d](#)[3d](#)[1w](#)[Custom](#) [📅](#)[UTC timezone](#) ▼[Explore related](#)[Configure CloudWatch agent](#)[Manage detailed monitoring](#)**CPU utilization (%)**

Percent

**Network in (bytes)**

Bytes

**Network out (bytes)**

Bytes

**Network packets in (count)**

Count



CPU utilization (%)



5 minutes ▼

Average ▼

1h

3h

12h

1d

3d

1w

Custom

UTC timezone ▼



Percent

50.9%

40.1

38.2%

25.5%

12.7%

0%

15:50 15:55 16:00 07-05 16:05 16:10 16:15 16:20 16:25 16:30 16:35 16:40 16:45 16:50

i-0dbba5d2558e8c59e (cloud-watch-SNS-110)

[View in metrics](#)

[Close](#)

When CPU Utilization graph cross the 40% then check email notification

Result

Compose

Inbox1,266

Starred

Snoozed

Sent

Drafts4

Purchases14

More

Labels+

Search mail

Upgrade

ALARM: "awsec2-i-0dbba5d2558e8c59e-GreaterThanOrEqualToThreshold-CPUUtili..." in US East (N. Virginia)

AWS Notifications<no-reply@sns.amazonaws.com>

22:19 (3 minutes ago)

You are receiving this email because your Amazon CloudWatch Alarm "awsec2-i-0dbba5d2558e8c59e-GreaterThanOrEqualToThreshold-CPUUtilization" in the US East (N. Virginia) region has entered the ALARM state, because "Threshold Crossed: 1 datapoint [50.93600068817777 (05/07/26 16:44:00)] was greater than or equal to the threshold (40.0)." at "Sunday 05 July, 2026 16:49:38 UTC".

View this alarm in the AWS Management Console:

<https://us-east-1.console.aws.amazon.com/cloudwatch/deeplink.js?region=us-east-1#alarmsV2:alarm/awsec2-i-0dbba5d2558e8c59e-GreaterThanOrEqualToThreshold-CPUUtilization>

Alarm Details:

- Name:

awsec2-i-0dbba5d2558e8c59e-GreaterThanOrEqualToThreshold-CPUUtilization

- Description:

Alarm on instance i-0dbba5d2558e8c59e: Triggered when CPUUtilization >= 40 for 1 consecutive 5-minute periods.

- State Change:

OK -> ALARM

- Reason for State Change:

Threshold Crossed: 1 datapoint [50.93600068817777 (05/07/26 16:44:00)] was greater than or equal to the threshold (40.0).

- Timestamp:

Sunday 05 July, 2026 16:49:38 UTC

- AWS Account:

667558012739

- Alarm Arn:

arn:aws:cloudwatch:us-east-1:667558012739:alarm:awsec2-i-0dbba5d2558e8c59e-GreaterThanOrEqualToThreshold-CPUUtilization

Threshold:

- The alarm is in the ALARM state when the metric is GreaterThanOrEqualToThreshold 40.0 for at least 1 of the last 1 period(s) of 300 seconds.

Monitored Metric:

- MetricNamespace:

AWS/EC2

- MetricName:

CPUUtilization

- Dimensions:

[InstanceId = i-0dbba5d2558e8c59e]

- Period:

300 seconds

- Statistic:

Average

- Unit:

not specified

State Change Actions:

- OK:

Reply

Forward